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RESEARCH ARTICLE

qTrustNet Virtual Private Network (VPN): Enhancing Security in the Quantum Era

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ABSTRACT This paper introduces qTrustNet VPN, a next-generation Virtual Private Network (VPN) designed to enhance security in the quantum era. As traditional encryption methods face obsolescence due to the advent of quantum computing, there is an urgent need for more robust security solutions. qTrustNet VPN addresses this need by ensuring the confidentiality and integrity of data transmission through the WireGuard protocol, known for its simplicity and high performance. At the core of qTrustNet VPN's enhanced security is the integration of a Post-Quantum Cryptography (PQC) algorithm, developed by the National Institute of Standards and Technology (NIST). This PQC algorithm is designed to withstand the decryption capabilities of quantum computers, thus providing long-term security. To further bolster security, the encryption key of the PQC algorithm is protected using Physically Unclonable Function (PUF) technology. PUFs offer a hardware-based security mechanism that is resistant to cloning and tampering, ensuring that the encryption keys remain secure against both classical and quantum attacks. This innovative combination of WireGuard protocol, PQC algorithms, and PUF technology creates a VPN system with significantly improved security features. To validate its effectiveness, the performance of the proposed qTrustNet VPN was rigorously evaluated in a 10GbE NIC network environment. The results demonstrated a remarkable throughput of up to 8.5 Gbps, ensuring high-speed data transmission capabilities. Additionally, the latency was measured at 1ms or less, highlighting the system's efficiency in maintaining low-latency communication. In summary, qTrustNet VPN represents a significant advancement in VPN technology, offering robust security features tailored for the quantum era while maintaining high performance and low latency. This paper details the design, implementation, and performance evaluation of qTrustNet VPN, showcasing its potential to secure data transmission against emerging quantum threats.

INDEX TERMS Physically unclonable function (PUF), post-quantum cryptography (PQC), virtual private network (VPN), WireGuard.

I. INTRODUCTION

With the rapid development of information and communication technology and the advent of the Internet of Things (IoT) era, several billions of IoT devices have been interconnected through networks [1]. These changes make our lives more convenient while also increasing the threat of personal information leakage through networks [2].

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Virtual Private Network (VPN) has been widely used to create a secure network environment. A Secure Sockets Layer (SSL) VPN is one of widespread kind of VPN due to strong security and high compatibility. However, complicated handshake procedures and high processing costs are incurred, settings are complicated, and maintenance is difficult. In addition, a high latency may occur, and performance in large-scale network environments is limited [3]. On the other hand, the WireGuard is simple and has excellent performance, but may be vulnerable to future security threats due

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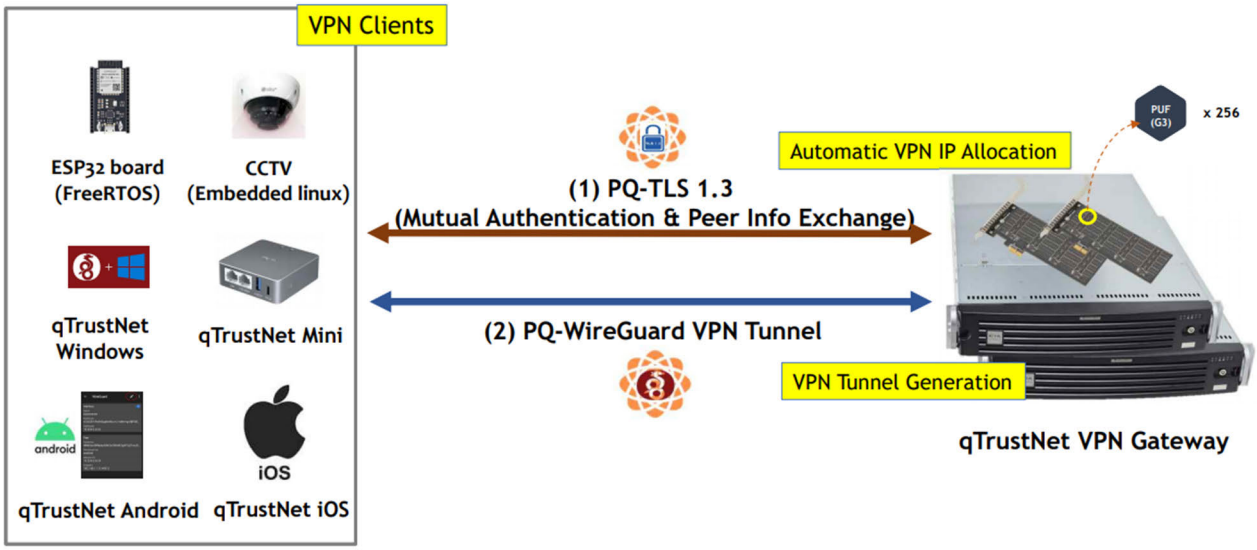

FIGURE 1. Basic structure and main protocol steps of qTrustNet VPN.

to a lack of resistance to quantum computing [4]. Quantum computers, with their unprecedented computational power, pose a significant threat to current encryption methods. Traditional asymmetric key encryption algorithms, such as RSA (Rivest-Shamir-Adleman) and ECC (Elliptic Curve Cryptography), rely on the mathematical complexity of problems like integer factorization and discrete logarithms for their security. However, quantum computers can solve these problems exponentially faster than classical computers. For instance, Shor's algorithm, running on a sufficiently powerful quantum computer, could break RSA and ECC encryption in polynomial time, rendering these encryption methods obsolete. In conclusion, the advent of quantum computing heralds a significant paradigm shift in cybersecurity [5], [6].

This paper proposes the qTrustNet VPN based on the WireGuard protocol for high throughput and low latency. Furthermore, the post-quantum cryptographic algorithms CRYSTALS Kyber and CRYSTALS Dilithium are implemented to achieve powerful security solution for the post quantum era. Furthermore, the secret key of the PQC algorithms is securely protected from physical attacks such as reverse engineering by physically unclonable function (PUF) technology. Due to its flexibility and effectiveness, the proposed qTrustNet VPN can be applied to various security applications such as intelligent closed-circuit television (CCTV), elevator CCTV, drone control, home networks, cold chain management, point-of-sale (POS) system, online shopping, remote medical services, public Wi-Fi and online gaming.

This paper is configured as follows. First, there is generally introduced how the qTrustNet VPN proposed in Section II is configured. In Sections III, IV, and V, WireGuard, PUF, and PQC that constitute the qTrustNet VPN will each be described.

II. OVERVIEW OF qTrustNet VPN

A. CONCEPT OF PROPOSED qTrust VPN

The qTrustNet VPN is an innovative virtual private network (VPN) solution that provides security against threats from the quantum era. This solution guarantees the confidentiality and integrity of data transmission by integrating a WireGuard protocol, a physically unclonable function (PUF) technology of an ICTK, and a quantum-resistant cryptography (PQC) algorithm of an NIST. [FIG. 1] shows a basic structure and main protocol steps of the qTrustNet VPN. This figure visually describes how the WireGuard protocol, the PUF technology, and the PQC algorithm are integrated. The system is composed of two stages: (1) the PQ-TLS 1.3 stage, which exchanges various information necessary for establishing a VPN tunnel by applying quantum-resistant algorithms to TLS 1.3, and (2) the PQ-WireGuard stage, which handles the actual creation of the VPN tunnel and the encryption/decryption of communication by applying quantum-resistant algorithms to WireGuard. This system supports VPN clients across various operating systems, including FreeRTOS, embedded Linux, Windows, macOS, Android, and iOS.

B. WireGuard PROTOCOL

The WireGuard is a simple, fast, and modern encryption protocol. The WireGuard is much simpler than existing IPsec or OpenVPN because it is composed of about 4000 lines of code, thereby reducing attack surfaces and making code review and auditing easier. In addition, The WireGuard guarantees the confidentiality and integrity of data using the latest encryption algorithm [7].

Table 1. illustrates the types of keys used, their respective algorithms, key sizes, and descriptions of their roles in the

TABLE 1. The types of keys used.

Key Type	Algorithm	Key Size	Description
ECDH Private Key	Curve25519	32 bytes	Private key used in the ECDH key exchange. Unique to each peer.
ECDH Public Key	Curve25519	32 bytes	Public key derived from the ECDH private key. Shared with the peer.
Kyber768 Private Key	Kyber768 (PQC)	2,400 bytes	Private key used for post-quantum secure key exchange.
Kyber768 Public Key	Kyber768 (PQC)	1,184 bytes	Public key used in the Kyber768 key exchange. Shared with the peer.
Kyber768 Ciphertext	Kyber768 (PQC)	1,088 bytes	Ciphertext generated during the Kyber768 key exchange. Sent to the peer.
Session Key	Derived (ECDH + Kyber768)	32 bytes	Final session key derived from ECDH and Kyber768 shared secrets. Used for encryption and authentication.

process. This addition provides a comprehensive view of how keys are generated, shared, and utilized during the qTrustNet VPN's secure key exchange.

C. PUF TECHNOLOGY

A Physical Unclonable Function (PUF) technology is a technology for generating unique identification information using the unique irregularities of physical structures. Most previous PUFs, e.g. SRAM, oscillator, or biased NAND, are based on active devices in the CMOS process. Because the electrical characteristics of the active device are sensitive to environmental conditions such as supply voltage and temperature, the PUF values based on the active device can be flipped according to the environmental conditions. Therefore, various preselection circuits with an error correction code are necessary to achieve high steadiness, which increase the processing time and latency. However, VIA PUF is based on passive devices such as a via, which is an interconnect layer between a silicon and a metal in a typical complementary metal-oxide semiconductor (CMOS) process [8], [9], [10]. Once the physical state of contact is determined after fabrication process, it can remain even in different operating conditions without any stability enhancement techniques. Therefore, the VIA PUF technology was applied to the proposed qTrustNet VPN, and this VIA PUF technology generates a unique encryption key for each device using the physical characteristics of a contact layer This provides high resistance to physical tampering and duplication with non-clonable characteristics, thereby guaranteeing the high level of security.

D. PQC ALGORITHM

E. A PUF technology can generate unique identification information based on minute process variation in the semiconductor fabrication process. The proposed qTrustNet VPN integrates VIA PUF technology, which utilizes contact formation probability to generate a unique secret key in a standard complementary metal-oxide semiconductor (CMOS) process [8], [9], [10]. VIA PUF ensures high stability and resistance to physical tampering, thereby guaranteeing a high level of security.

III. MAIN FUNCTIONS OF qTrustNet VPN

A. MUTUAL AUTHENTICATION

1) PUF-BASED AUTHENTICATION

Using the PUF technology, PUF values generated using minute irregularities that occur in a semiconductor manufacturing process are encrypted using private keys or security keys.

2) AUTHENTICATION PROCESS AND STEPS

In the authentication step, a PQ-TLS 1.3 protocol is used. The PQ-TLS1.3 protocol is a protocol combining TLS 1.3 with the PQC algorithm (CRYSTALS Kyber, CRYSTALS Dilithium), and Kyber KEM (key exchange) was integrated in a TLS 1.3 Client Hello & Server Hello step, and Dilithium-based x.509 signature/verification processes were integrated in a certificate delivery and verification process. Then, the counterpart's VPN information required for a WireGuard step is exchanged between a client and a server and the reliability of the client is verified, and all subsequent processes are encrypted using the private key previously exchanged through the Kyber KEM.

Describing a process of exchanging the VPN information, the client transmits, to the server, information, such as a MAC address, a serial number (SN), and a public key. After the server verifies the reliability of the client based on this information, the server transmits, to the client, its VPN information and client ID, which have been generated based on the PUF. Through this process, the client and the server finish mutual authentication and exchange WireGuard keys for safe communication. [FIG. 2] describes a mutual authentication step between the client and the server. This process is safely performed through the PUF-based authentication and the PQ-TLS 1.3 protocol.

Examining the process of exchanging VPN information, the client sends information such as the MAC address, serial number (SN), and public key to the server. Based on this information, the server verifies the reliability of the client. After verification, the server sends its VPN information, generated based on PUF, along with the client ID, to the client. Through this process, the client and server complete mutual authentication and perform a WireGuard key exchange for secure communication. [FIG. 3] shows the process of exchanging VPN information between the client and server.

B. EXCHANGE WireGuard KEY

1) CRYSTALS KYBER ALGORITHM

The CRYSTALS Kyber is a public key encryption algorithm and provides safe key exchange even against attacks from

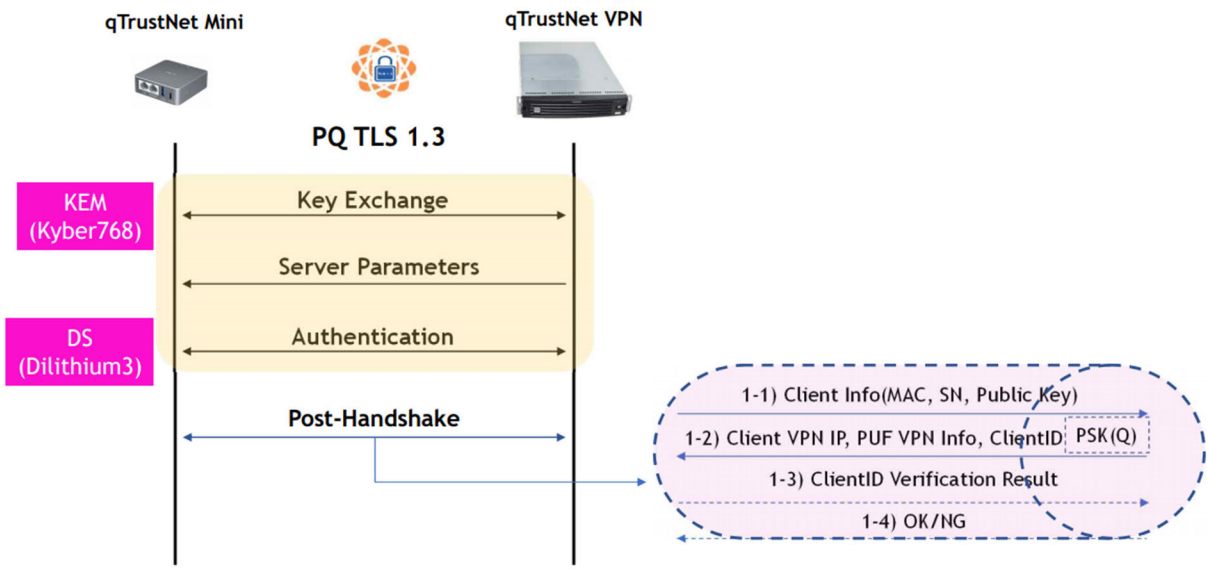

FIGURE 2. Mutual authentication step between the client and the server.

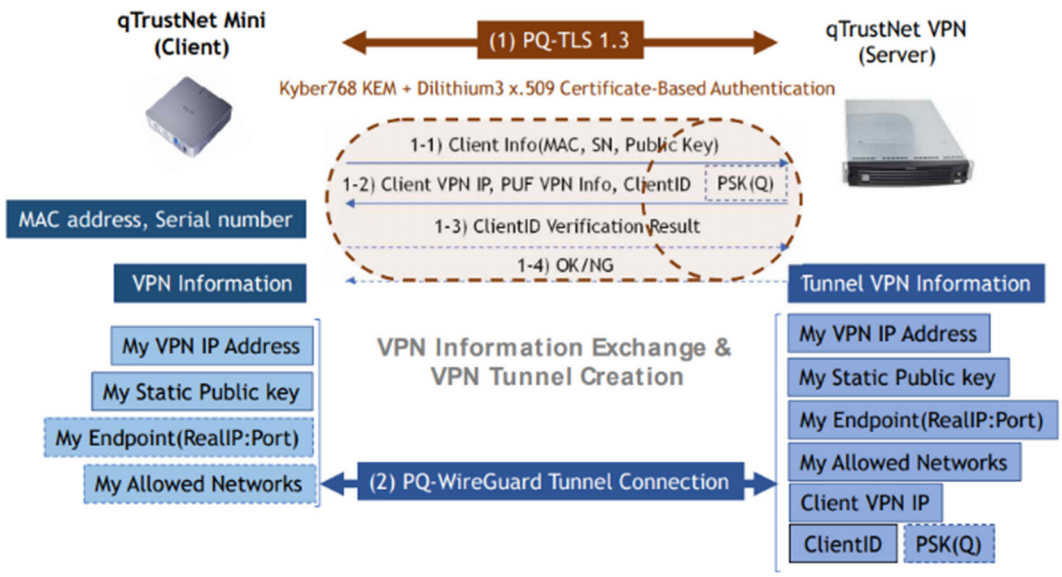

FIGURE 3. VPN information exchange stage between client and server.

quantum computers. This algorithm provides high security and efficiency and is stably operated in various network environments [12].

2) KEY EXCHANGE PROCEDURE

The key exchange procedure begins with the client and the server each generating a pair of a public key and a private key. The client transmits its public key to the server, and the server encrypts a session key using the public key of the client and then transmits the encrypted session key to the client. The client obtains the session key by decrypting the encrypted session key received from the server using its private key. Therefore, both parties share a common session key for safe communication. Through this process, the client and the server exchange the session keys that may guarantee safe communication even against attacks from quantum computers. [FIG. 4] shows the key exchange method of the WireGuard protocol, while [FIG.5] shows the hybrid key exchange procedure that combines the WireGuard protocol with the CRYSTALS Kyber algorithm. WireGuard performs four instances of ECDH (Elliptic-curve Diffie-Hellman) to exchange symmetric keys between the VPN client and server.

By integrating the CRYSTALS Kyber algorithm, an additional key exchange is conducted, ultimately exchanging a quantum-resistant symmetric key.

C. DATA TRANSMISSION THROUGH ENCRYPTED TUNNEL

1) ENCRYPTION METHOD OF WireGuard PROTOCOL

The WireGuard guarantees the confidentiality and integrity of data using simple and powerful encryption algorithms (curve25519, blake2s, chacha20, poly1305). This protocol maintains high security during data transmission using the latest encryption technology.

2) TUNNEL SETUP AND DATA TRANSMISSION PROCEDURE

The encrypted tunnel setting and data transmission procedure begins when the key exchange is finished. When a VPN tunnel is setup, all pieces of data are transmitted in an encrypted state through this tunnel. Data in the tunnel is protected by applying the latest encryption algorithm through the WireGuard protocol, and communication security is continuously enhanced through periodic key exchange. This process guarantees the confidentiality and integrity of data and is essential to provide a safe communication environment.

D. SUPPORT ON VARIOUS OPERATING SYSTEMS

Support is available on operating systems such as FreeRTOS, Embedded Linux, Windows, Mac OS, Android, and iOS. Such a broad operating system support allows the qTrustNet VPN to be used on various devices, thereby greatly increasing the flexibility and convenience of security communication. For example, a user may safely communicate through the qTrustNet VPN on various devices, such as desktop computers, smartphones, and tablets. This acts as an important advantage not only for individual users but also in corporate environments. Enterprises may adopt consistent security policies using the qTrustNet VPN when connecting different types of devices using various operating systems to the network.

IV. ENHANCED SECURITY THROUGH WireGuard

A. SECURITY CHARACTERISTICS OF WireGuard

A WireGuard design facilitates code review and auditing and enhances security by reducing the attack surfaces. Such a characteristic makes the WireGuard a powerful and efficient VPN solution. [FIG. 6] describes a basic concept and advantages of the WireGuard protocol.

B. PACKET PROCESSING PROCESS

The packet processing process is optimized to safely manage data transmission between the client and the server, and each packet is protected using a unique encryption key. In this process, all pieces of data are transmitted in an encrypted state and is decrypted only at a receiving side. Therefore, it is possible to maintain the confidentiality and integrity of data and minimize the possibility of data theft or tampering during transmission.

C. COMPARISON BETWEEN WireGuard AND TLS

The WireGuard provides various advantages in terms of performance compared to the TLS [13]. The WireGuard sets a connection using minimal round-trip communication through a simple initial connection setting process without a complicated SSL/TLS handshake process. In addition, stronger authentication can be provided through the unique physical characteristics-based mutual authentication using the PUF technology. Quantum-safe key exchange is performed using the CRYSTALS Kyber algorithm, and high security is maintained through efficient key exchange. These characteristics mean that the WireGuard provides a simpler and more efficient setting process, stronger mutual authentication, lower overhead, and higher performance than the TLS.

D. VPN PERFORMANCE COMPARISON

As the use of virtual private networks (VPNs) has recently increased, interest in comparing the performance of various VPN protocols is also increasing. Representative VPN protocols include IPSec, OpenVPN, and WireGuard, and each has advantages and disadvantages. The IPSec is a standard VPN protocol with a long history and provides high security and stability. However, the IPSec has a disadvantage of being slightly slower and having more complicated settings that other protocols [14]. The OpenVPN is an open source VPN protocol and provides faster speeds and more flexible settings than the IPSec, but has a disadvantage of weak security and complicated settings option [15]. The WireGuard is the latest VPN protocol and provides a high speed, simple settings, and excellent security at the same time. The WireGuard supports limited platforms and has a disadvantage of low stability [16]. The qTrustNet VPN combines the PQC algorithm with the existing WireGuard protocol and safely stores the WireGuard private key using the PUF, thereby increasing security. In addition, the qTrustNet VPN is optimized and ported for each of the operating systems, such as FreeRTOS, embedded Linux, Windows, Mac OS, Android, and iOS, to complement the existing WireGuard's disadvantage of limited platform support. Such contents are summarized in a Table 1.

E. PERFORMANCE FACTORS

The WireGuard protocol enables low latency and high throughput and can maintain high performance even in environments where low-power devices or resources are limited. The WireGuard design has minimized code, thereby reducing attack surfaces and enhancing security. In addition, it is possible to minimize latency and maximize data transmission speed by efficiently optimizing network traffic. This is especially useful in applications where data transmission is critical, such as high-definition video streaming or real-time communication. Furthermore, this may also be applied to encrypted connections between sites, potentially replacing dedicated lines.

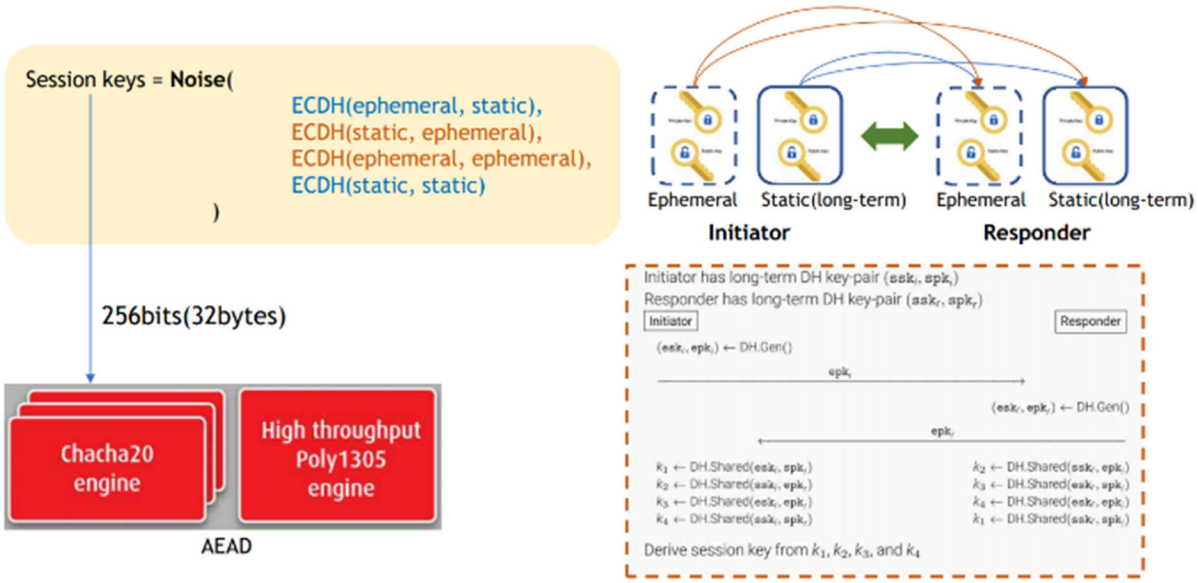

FIGURE 4. Original WireGuard key exchange procedure.

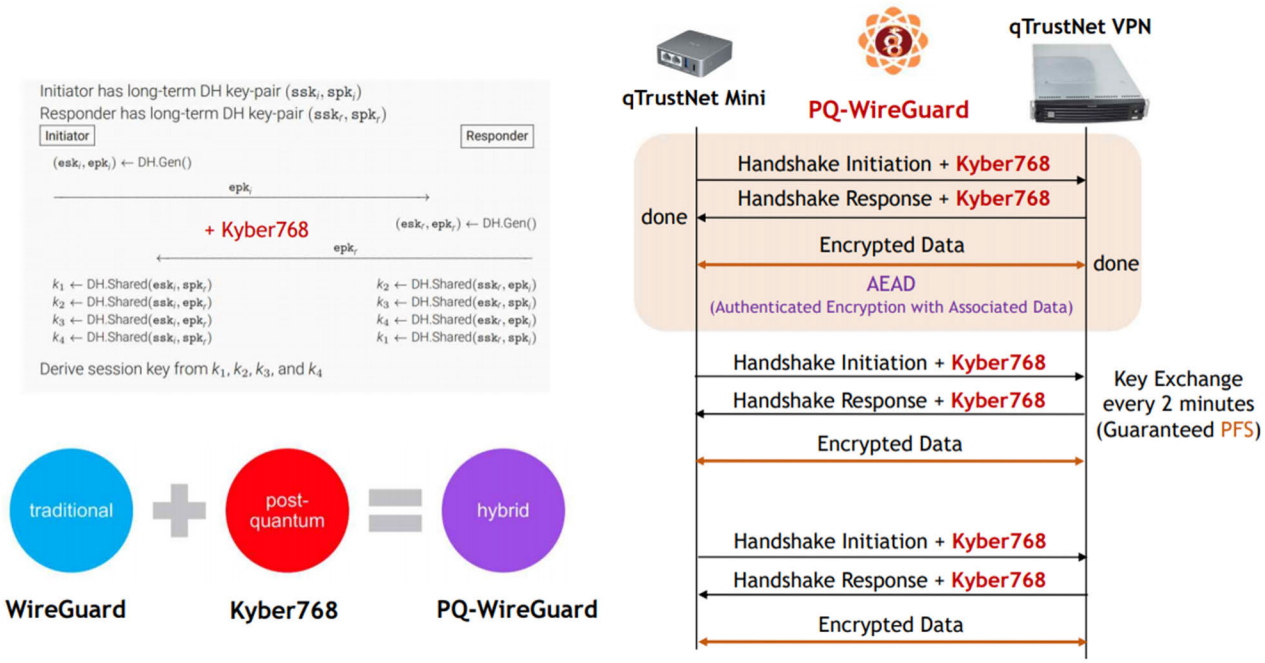

FIGURE 5. Key exchange procedure using CRYSTALS Kyber algorithm.

F. REAL-TIME PERFORMANCE TEST

It was confirmed that the performance of the qTrustNet VPN was tested in various environments and a high throughput and a low latency were maintained. For example, in a testbed based on an Intel Xeon processor, the qTrustNet VPN based on the WireGuard supported up to 5,000 simultaneous connections and exhibited stable data transmission rates. Such performance demonstrates that the qTrustNet VPN may provide stable security even in large-scale network environments. Real-time performance test results are an important indicator for confirming the efficiency and reliability of the qTrustNet VPN.

V. PUF TECHNOLOGY INTEGRATION

A. PRINCIPLE AND ADVANTAGES OF PUF

The fundamental principle of a PUF is to generate unique identification information for a semiconductor chip from its process variations. Since the PUF technology generates the unique identification information inside the semiconductor chip, it can inherently prevent information leakage problems

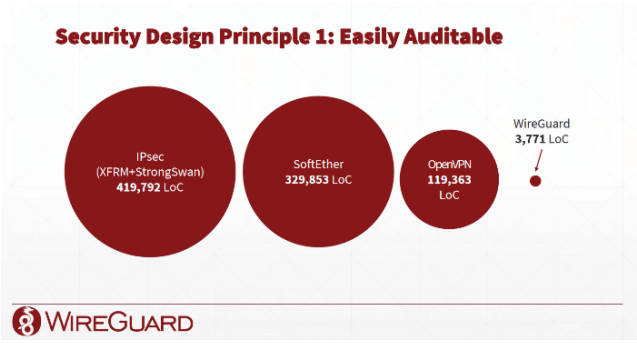

FIGURE 6. Basic concept of WireGuard protocol.

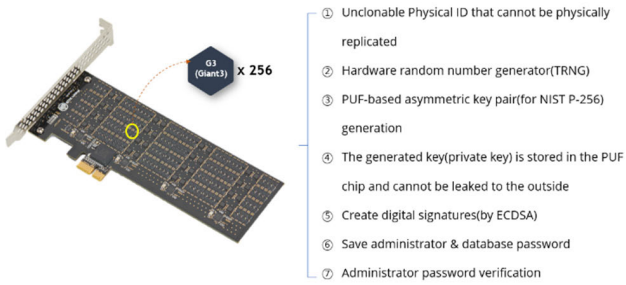

FIGURE 7. PUF Array (qTrustPCI) equipped with 256 G3 PUF chips.

that can occur when unique information is generated outside the semiconductor chip and injected into a non-volatile memory inside the chip [19].

B. PUF INTEGRATION IN qTrustNet VPN

The qTrustNet VPN performs device uniqueness authentication using the PUF technology. Each VPN client performs the device uniqueness authentication using the security key generated based on the PUF, and the server confirms whether the client is a reliable device by verifying the client's PUF response. Through the PUF-based authentication, a reliable relationship between the client and the server are constructed, and safe communication is guaranteed. [FIG. 7] depicts the PUF Array (qTrustPCI) equipped with 256 G3 PUF chips and highlights its main functions. [FIG. 8] shows the PUF Array installed in a PCI slot within the qTrustNet VPN equipment.

C. TECHNICAL IMPLEMENTATION

The technical implementation is realized by generating the PUF value using the PUF technology, then encrypting the private key or security key used in the proposed qTrustNet VPN using the generated PUF value, and storing the encrypted private key or security key in a storage included in the VPN. The generated security key is safely stored through the PUF and used for encrypted communication as needed.

The G3 PUF chip is based on VIA PUF technology. The contact is a kind of via layer used to connect the silicon and the first metal layer, designed with specific physical

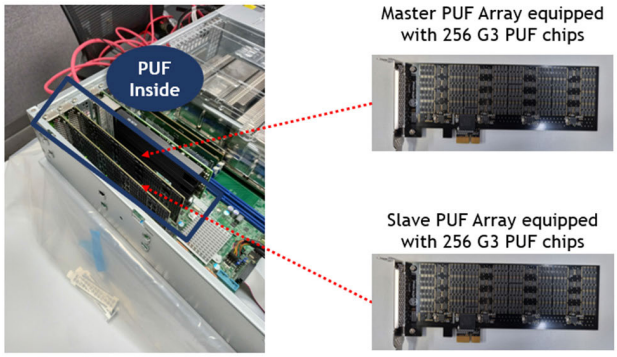

FIGURE 8. PUF Array installed in a PCI slot.

dimensions to ensure connectivity. However, if the contact hole size is smaller than the design rule, it may remain open after the fabrication process [8], [9], [10]. As the contact hole size decreases, the probability of successful contact formation also decreases. Contact formation can become stochastic within a certain range of contact hole sizes. Furthemore, this stochastic range can vary due to die-to-die, wafer-to-wafer, lot-to-lot, and run-to-run process variations. Therefore, the G3 PUF chip has a large number of PUF bitcells with various contact hole size, and each bitcell incorporates four contacts to reduce the area. These PUF bitcells are designed according to the physical structure of the standard cell library and placed randomly with other logic gates within the chip to enhance tamper resistance to physical attacks, such as reverse engineering or optical analysis.

D. APPLICATIONS AND CASE STUDIES

The PUF technology is proving usefulness in various security applications. For example, the PUF technology enables real-time safe data transmission in intelligent CCTV security, drone control security, cold chain security, remote medical security, etc. [19].

E. PERFORMANCE EVALUATION

It was confirmed that the performance of the PUF technology was tested in various environments and the PUF technology provided the high level of security. The qTrustNet VPN verifies the reliability of each device through the PUF-based device uniqueness authentication and safe key management and maintains the high level of security even in the quantum era.

VI. POST QUANTUM CRYPTOGRAPHY (PQC) INTEGRATION

A. NECESSITY OF PQC

Since existing encryption algorithms such as RSA and ECC may be easily decrypted by quantum computers, there is a need for a quantum-safe encryption algorithm. The PQC algorithm provides a safe encryption technology even against attacks from quantum computers [20].

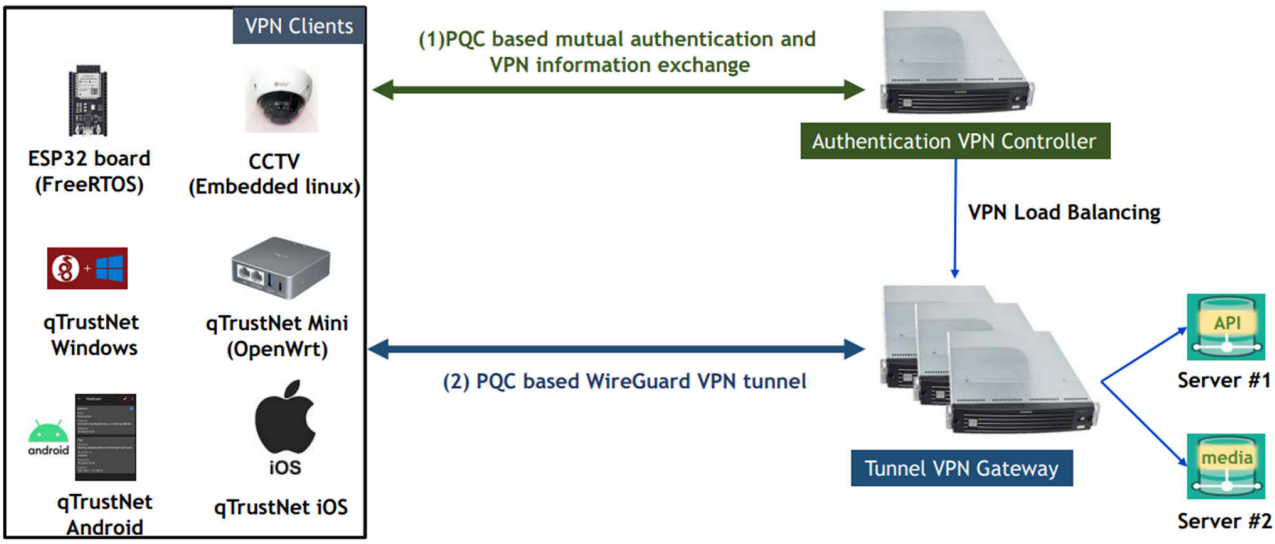

FIGURE 9. PQC and PUF integration in qTrustNet VPN.

B. CRYSTALS KYBER AND CRYSTALS DILITHIUM

The CRYSTALS Kyber is a public key encryption algorithm and guarantees safe key exchange even against attacks from quantum computers, and the CRYSTALS Dilithium is a digital signature algorithm and has fast and safe signature generation and verification [21].

C. PQC AND PUF INTEGRATION IN qTrustNet VPN

The structure of the above-described qTrustNet VPN is a structure that processes both authentication and tunneling in one device, but as shown in [FIG. 9] below, may be configured as a structure of an authentication VPN (Controller) serving as a Software Defined Perimeter (SDP) or a tunnel VPN (Gateway) performing tunnel connection with the client through client authentication and VPN session management. Using the CRYSTALS Kyber and CRYSTALS Dilithium algorithms, quantum safe key exchange is performed between the authentication VPN server and the client through the PQC key distribution (Key Encapsulation Mechanism (KEY)), and mutual authentication is guaranteed through digital signatures. Then, safe tunneling is guaranteed in a hybrid manner by exchanging VPN information through a quantum safe communication channel and coupling the PQC key distribution (Key Encapsulation Mechanism (KEY)) to the existing ECDH key exchange during WireGuard Handshake for tunnel connection between the tunnel VPN server and the client. Since the WireGuard Static Private Key in an H/W RoT chip is safely stored in the server, the corresponding key may be safely managed in the server even without hardware support of a terminal. In addition, access request is processed through the authentication VPN server, and the client's identity is verified. In addition, safe data transmission is guaranteed by checking the VPN traffic of each tunnel VPN server in real time, assigning an optimal tunnel VPN server to the client through load balancing, managing VPN session settings.

TABLE 2. Virtual VPN performance comparison [17].

	IPSec VPN	OpenVPN	WireGuard	qTrustNet VPN
Rate Mb/s	2,333	495	4,489	4,489
Security	High	Normal	High	High
Stability	High	Normal	Normal	High
Settings	Complicated	Normal	Simple	Simple
Platform support	Broad	Broad	Limited	Broad

D. PERFORMANCE EVALUATION

It was confirmed that the performance of the PQC algorithm was tested in various environments, and the PQC algorithm provided the high level of security and at the same time, was operated efficiently. The qTrustNet VPN can safely protect data against threats from quantum computers by integrating the CRYSTALS Kyber and Dilithium algorithms.

VII. PERFORMANCE EVALUATION

A. TEST ENVIRONMENT

Performance evaluation was conducted on the testbed based on the Intel Xeon processor. Network settings were performed in a high-speed network environment using 10GbE NIC, and qTrustNet VPN software based on the WireGuard protocol was applied. Table 2 shows the hardware specifications of the device used for actual performance testing. [FIG. 10] depict the test environment.

B. THROUGHPUT

The qTrustNet provides a lightweight and faster performance compared to existing IPSec or OpenVPN solutions, primarily because it operates at the kernel level. Additionally, it utilizes user datagram protocol (UDP), which imposes less load on the server compared to the connection-oriented protocols like TCP that require more resources. For these reasons,

TABLE 3. Hardware specifications of qTrustNet VPN.

Item	Specification	Note
Bay Slot	2U	
CPU	Intel® Xeon® Processor Gold 5218 16 core x 2(dual socket), 2.3GHz	Total 64 CPU threads
RAM	16GB x 4	Total 64GB
SSD	480GB x 2	OS storage
Storage (HDD)	2TB x 2	Log storage
Ethernet Ports	1GbE RJ45 2 ports + 10GbE RJ45 4 ports	Integrated VPN
PUF Array	PUF Array(TrustPCI) x 2	2 x 256 G3 chips PCIe 2.0 x1
Serial Interface	COM Port 1	RS232
Power	1+1 1U 800W Redundant Power Supply 76mm width, W/PMbus, Titanium	

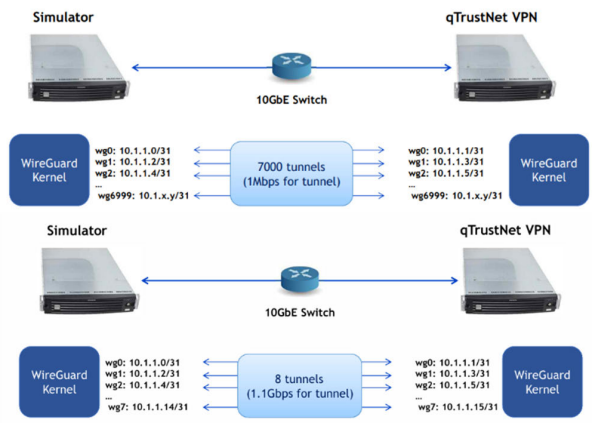

FIGURE 10. Test environment in case of (a) 7000 tunnels with 1 Mbps for tunnel, and (b) 8 tunnels with 1.1 Gbps for tunnel.

qTrustNet has a performance advantage over traditional VPN protocols such as OpenVPN or IPSec.

Performance evaluation was conducted on the testbed based on the Intel Xeon processor. Network settings were performed in the high-speed network environment using 10GbE NIC, and the throughput of up to 10 Gbps was recorded using WireGuard 256 bits ChaCha20 and Poly1305 encryption. [FIG. 11] shows that average per qTrustNet VPN tunnel Mbit/sec. We tested the qTrustNet VPN under conditions with up to 7000 tunnels, with each tunnel measured at 1 Mbps throughput. Even under such load conditions, the qTrustNet VPN maintained stable tunnel performance without any significant degradation. This demonstrates that the proposed solution can support a high number of connections while minimizing overhead. [FIG. 12] presents the average throughput of the qTrustNet VPN over a 1-hour period (3600 seconds), during which a maximum throughput of 8.5 Gbps was observed. This result highlights the system's capability to sustain large-scale data transfers over extended periods, further showcasing the efficiency and robustness of the proposed solution in comparison to traditional VPNs. [FIG. 13] shows that throughout the testing process, system

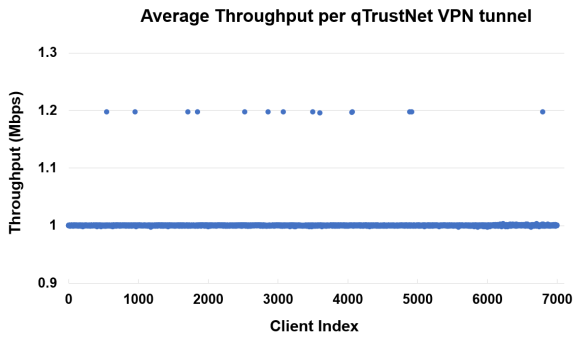

FIGURE 11. Test result 1.

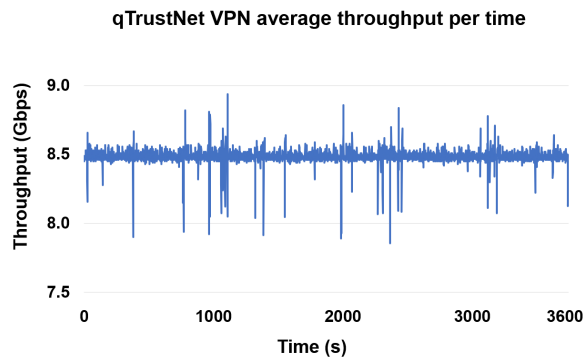

FIGURE 12. Test result 2: Throughput.

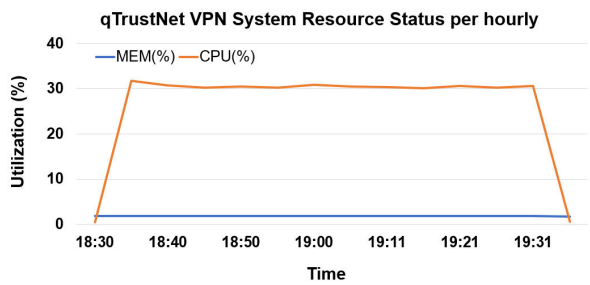

FIGURE 13. Test result 3: qTrusNet VPN system resource status.

stability was maintained, with CPU usage peaking at 30% and memory usage reaching a maximum of 2%. These metrics indicate that the qTrustNet VPN is capable of maintaining high throughput while efficiently utilizing system resources, even in large-scale data transmission environments.

qTrustNet VPN is an optimized solution for enterprise and high-bandwidth applications, offering a robust balance between security, performance, and resource efficiency. While its resource requirements are marginally higher than WireGuard, qTrustNet compensates with significantly enhanced security features, including Perfect Forward Secrecy (PFS) and advanced Post-Quantum Security measures, making it resilient to both current and future threats such as quantum computing. Additionally, qTrustNet achieves comparable throughput and connection speeds, with efficient handling of large packet sizes, while also providing

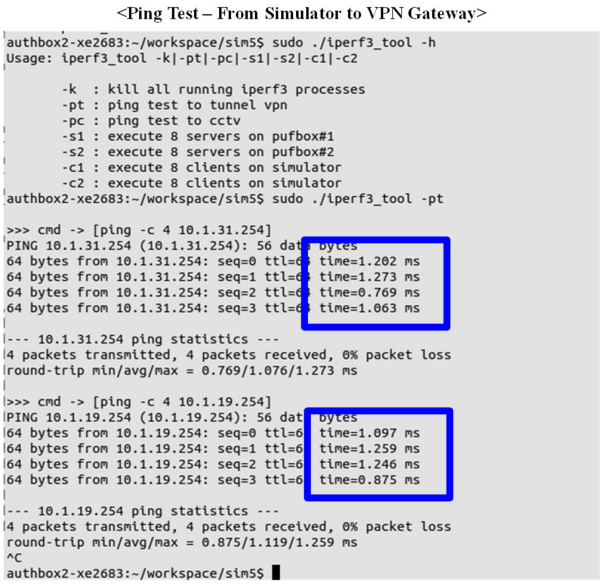

FIGURE 14. Test result 4: Ping test (Latency).

TABLE 4. Comparison of qTrustNet VPN with widely used solutions.

Meric	WireGuard	qTrusNet VPN	Open VPN	IPSec
Handshake Time	1~2ms	3~5ms	10~20ms	5~15ms
Throughput	800~900 Mbps	700~800 Mbps	400~600 Mbps	600~800 Mbps
CPU Usage	≤ 5%	10~15%	20~30%	15~25%
Memory Usage	≤ 10MB	15~20MB	20~30MB	25~35MB
Packet Size	96 bytes	2,400~2,600 bytes	400~500 bytes	500~700 bytes
Security (PFS)	PFS Supported	PFS + Post-Quantum Security	Limited PFS Support	Limited PFS Support
Ease of Setup & Management	Very Easy	Easy	Moderately Difficult	Difficult

easier management compared to OpenVPN. These attributes position qTrustNet VPN as a highly attractive and reliable choice for environments requiring high levels of security and data transmission efficiency.

C. LATENCY

As a result of the latency test, the qTrustNet VPN maintained an average latency of 1ms or less [FIG. 14]. This is a very important factor in real-time applications, such as video streaming or online games. The optimized network traffic management and efficient encryption process of the qTrustNet VPN enabled a low latency. Furthermore, the qTrustNet VPN, successfully handled up to 8 tunnels, each carrying a load of 1.1 Gbps, while maintaining an average latency of less than 1 ms. This result proves that even with the integration of Post-Quantum Cryptography (PQC) and Physically Unclonable Function (PUF) technologies, the impact on latency is minimal in high-performance network environments?. Built on the lightweight design of WireGuard, qTrustNet VPN has demonstrated negligible performance degradation despite the added quantum-resistant security features, outperforming other existing VPN solutions such as OpenVPN and IPSec.

D. CONNECTION STABILITY

Up to 7,000 simultaneous connections are supported, and high connection stability is maintained. Stable connection was provided despite connection attempts from various client devices, and an operation was stably performed without performance degradation even when the connection was maintained for a long time.

E. SECURITY

All current VPN servers face the challenge of protecting the key encryption key (KEK). Traditional methods to protect KEK involve not storing the key or generating it through password combinations. However, using PUF technology and encrypted secure storage based on PUF to store KEK is a highly advanced idea from a security perspective. This method of storing KEK in PUF-based secure storage not only enhances qTrustNet's security but could also improve the security of existing VPN solutions like IPSec and OpenVPN.

Decrypting qTrustNet requires intercepting the private key during the initial key distribution. However, the implementation of Forward Perfect Security (FPS) dynamically changes symmetric keys every 20 seconds, making consistent interception highly resilient. Moreover, PUF keys inherently possess immutable properties. The PUF chip used has undergone rigorous penetration testing and has been validated through Common Criteria (CC) certification. Simulated attacks require modeling challenge-response pairs, but the use of unpredictable session keys generated by a True Random Number Generator (TRNG) makes such simulations practically infeasible.

F. USABILITY

The private key of each VPN client is not stored on the server; instead, the server manages a password for client authentication. There is no need to store the client's private key on the server; the client should keep it encrypted, while the server only holds the client's public key. This technique, used by the proposed qTrustNet VPN, can be the best solution for devices that cannot immediately implement hardware-based security. This architecture is a unique feature of the qTrustNet VPN, which can also be applied to existing VPNs like OpenVPN or IPSec. The qTrustNet VPN's method for protecting the key on the client, where the VPN client is installed, and on the VPN server uses PUF (Physical Unclonable Function) as a key security enhancement technique. It is a universal technology that can be applied to other VPN technologies as well.

G. SUMMARY OF PERFORMANCE TEST

It was confirmed that the performance was tested in various environments and the high throughput and the low latency were maintained. Such performance demonstrates that stable

security may be provided even in large-scale network environments.

VIII. APPLICATIONS

The proposed qTrustNet VPN is fundamentally a solution designed for environments requiring high bandwidth. It targets enterprise environments such as CCTV systems, where, for example, approximately 5,000 concurrent CCTV cameras continuously generate around 2Mbps of traffic each. For this reason, it is more suitable for environments demanding large amounts of bandwidth. For instance, instead of using a dedicated line for secure communication between a company's headquarters and branch offices, the qTrustNet VPN can provide an equivalent or higher level of security at a more cost-effective price. According to the U.S. CNSA 2.0 transition plan, all networks must support PQC (Post-Quantum Cryptography) by 2033. Therefore, existing networks based on asymmetric key algorithms like ECC and RSA must be upgraded to support PQC. The proposed qTrustNet VPN is well-suited for such PQC transitions and is expected to be widely used in secure environments such as VPNs and cloud connectivity in the future.

The high-performance VPN and wide-bandwidth VPN server are specifically designed for enterprise applications but are also well-positioned to expand into personal VPN use cases. The enterprise applications are optimized to support concurrent 2Mbps video streams, while personal users, though requiring lower bandwidth per connection, typically involve a larger number of simultaneous connections. To address this, future plans include expanding to IPv6, enabling the provision of significantly more IP addresses to support broader scalability.

A. 8.1. INTELLIGENT CCTV SECURITY

The qTrustNet VPN guarantees the integrity of CCTV devices and prevents unauthorized access by providing safe video streaming in real time from intelligent CCTV systems. Therefore, the reliability of the CCTV can be secured, and video data can be transmitted safely.

B. ELEVATOR CCTV SECURITY

Safety inside elevators is enhanced by providing reliable real-time monitoring of elevator CCTVs through qTrustNet VPN relay. Unauthorized access and data leakage are prevented by safely transmitting elevator CCTV data.

C. DRONE CONTROL SECURITY

Unauthorized access and data leakage are prevented by safely transmitting control data and video data of drones through the qTrustNet VPN. Therefore, it is possible to enhance the security of operations and data transmission of the drones.

D. HOME NETWORK SECURITY

Safe communication is guaranteed by logically separating home networks, and safe connection of each device is maintained through the qTrustNet VPN. Therefore, the integrity of personal information and data are protected by enhancing home network security.

E. COLD CHAIN SECURITY

An IoT cold chain system maintains and safely manages the freshness of food by monitoring and controlling temperature and humidity states in real time. Therefore, unauthorized access to cold chain data is prevented, and data confidentiality is guaranteed.

F. POS SECURITY

Data leakage during the payment process is prevented by safely transmitting payment data between a POS terminal and a VAN server. Therefore, a safe payment environment is provided.

G. ONLINE SHOPPING SECURITY

By enhancing the security of online transactions, safe transactions between users and sellers are guaranteed, and personal and payment information are protected. Therefore, unauthorized access and leakage of data when shopping online are prevented.

H. REMOTE MEDICAL SECURITY

A remote medical system protects patient personal information through the qTrustNet VPN and supports safe remote medical treatment and video conferencing. Therefore, the confidentiality and integrity of medical data are guaranteed, and safe remote medical services are provided.

I. PUBLIC Wi-Fi SECURITY

By automatically attempting VPN connection when connecting to public Wi-Fi, safe connection is guaranteed, and security threats when using public networks are minimized. Therefore, the confidentiality of data when using public Wi-Fi is maintained, and a safe network environment is provided.

J. ONLINE GAME SECURITY

By providing a safe online gaming environment without changing firewall or router settings, game data is protected, and smooth gaming experiences are provided. Therefore, unauthorized access and leakage of online game data are prevented, and safe game play is guaranteed.

IX. POTENTIAL LIMITATIONS AND CHALLENGES

The proposed qTrustNet is based on the WireGuard protocol to provide VPN services. While existing protocols like IPSec and OpenVPN have been standardized by international organizations such as IETF and ISO, WireGuard has yet to achieve the same international standardization status, which raises concerns about the scalability of qTrustNet. Consequently, many clients request protocols that are compatible with widely adopted standards like IPSec or OpenVPN, and this demand is understandable to a certain extent.

However, qTrustNet is already supported by the latest versions of Android, and companies like IP Times have also adopted WireGuard as a default VPN protocol due to its lightweight nature. This indicates that WireGuard is gradually gaining more adoption and scalability in terms of usability. Performance-wise, WireGuard, the foundational protocol for qTrustNet, inherently offers significant advantages. For instance, in its application to CCTV systems, qTrustNet has demonstrated superior performance.

Therefore, while qTrustNet may face some scalability challenges in certifications like Common Criteria (CC), it is expected that future expansions of CC certifications will allow for broader protocol scalability. As PQC (Post-Quantum Cryptography) does not yet have international certification organizations, it is anticipated that future standards such as FIPS 140-3, including CAVP/CMVP, will integrate PQC. These developments in standardization and certification hold great potential for expanding qTrustNet's scalability.

The performance superiority of qTrustNet is expected to remain a key advantage in the future.

Furthermore, the client was primarily tested in IoT environments with constrained resources, such as CCTV systems. Additionally, the latest Android smartphones support the WireGuard protocol at the kernel level, confirming its suitability for performance optimization. Specifically, the WireGuard protocol, being lightweight, has been validated to function effectively in embedded environments and resource-constrained IoT scenarios.

X. CONCLUSION

As a result of the performance evaluation, it was confirmed that the qTrustNet VPN maintained high throughput and low latency while operating stably in various applications. This study demonstrated the effectiveness and efficiency of the qTrustNet VPN as a powerful security solution against threats from quantum computers. The qTrustNet VPN is differentiated from existing VPN solutions through various technical contributions. First, by integrating the PUF technology, the uniqueness of each device is authenticated, and a high level of security is maintained. Second, with the post-quantum encryption integration through the CRYSTALS Kyber and CRYSTALS Dilithium algorithms, key exchange and digital signatures were implemented even against attacks from quantum computers. Third, high performance and security are simultaneously provided based on the simple and efficient design of the WireGuard protocol. Additionally, practical challenges in deploying the qTrustNet VPN in heterogeneous environments, such as legacy systems and resource-constrained devices, were carefully considered. The lightweight design of the WireGuard protocol, coupled with the energy-efficient implementation of PQC and PUF technologies, enables compatibility with low-power IoT devices while maintaining robust security. Strategies such as adaptive scaling of cryptographic operations and leveraging modular hardware support can further facilitate integration with diverse infrastructure, including older or limited-capacity systems.

Future research may be conducted in the following directions to further improve the performance and security of the qTrustNet VPN. First, the introduction of hardware-based acceleration technology could enhance the performance of encryption and decryption processes. Currently, PQC algorithms have been implemented in software. However, following the NIST PQC announcement in August 2024, it has become increasingly feasible to implement PQC algorithms in hardware. This approach can offer benefits in speed through hardware acceleration, allowing performance improvements. Second, developing advanced network optimization algorithms for maximizing efficiency in large-scale network environments is necessary. These algorithms could address scalability challenges in deployments requiring thousands of simultaneous connections, such as global enterprise systems. Third, there is a need for research to increase the accessibility and usability of the qTrustNet VPN by improving user interfaces and adapting to various applications. For instance, integrating AI-driven threat detection can enhance the VPN's ability to identify and respond to emerging threats in real-time. Additionally, federated learning techniques can bolster security while preserving user privacy by enabling collaborative security model training across distributed devices. Furthermore, utilizing next-generation high-speed security chips with PCI security modules can contribute to performance improvements, ensuring that the qTrustNet VPN remains an adaptable and future-proof solution for the evolving cybersecurity landscape.

The qTrustNet VPN is a powerful solution capable of addressing the rapidly increasing security threats posed by quantum computers. The integration of the WireGuard protocol, PUF technology, and PQC algorithms has demonstrated its high level of security, as evidenced through performance evaluations and analyses across various applications. The qTrustNet VPN is poised to become an essential solution, meeting diverse current and future security requirements. This study highlights the technical contributions and potential of the qTrustNet VPN and evaluates its position as a critical security solution in the post-quantum era.

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